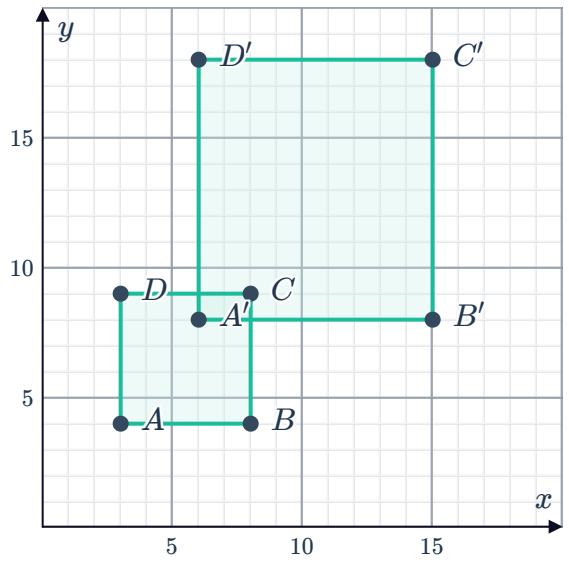
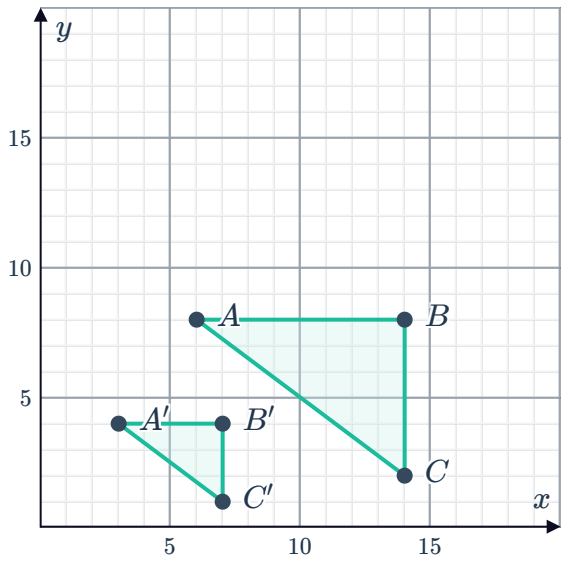
Dilations

1 Determine whether the following graphs show a dilation. If yes, find the scale factor.

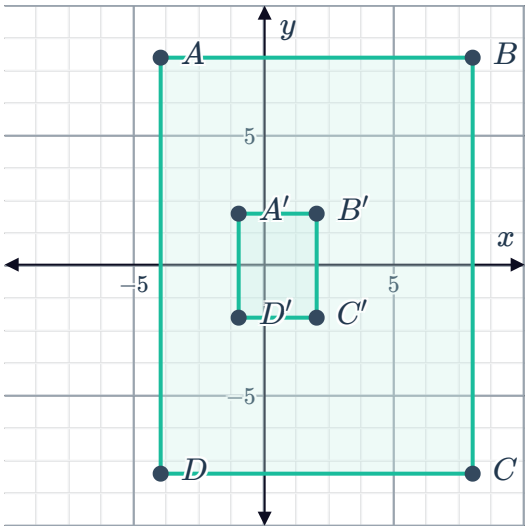
a



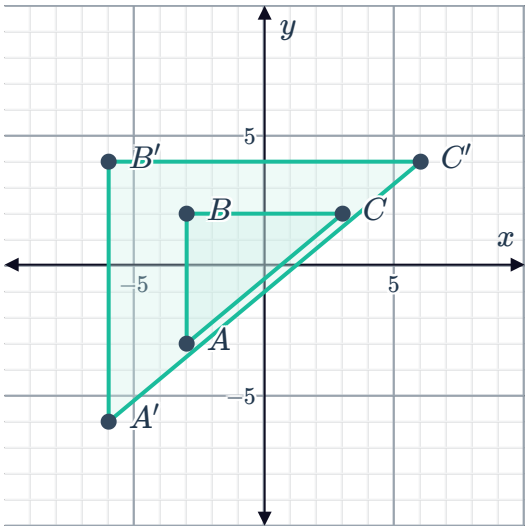
b



c



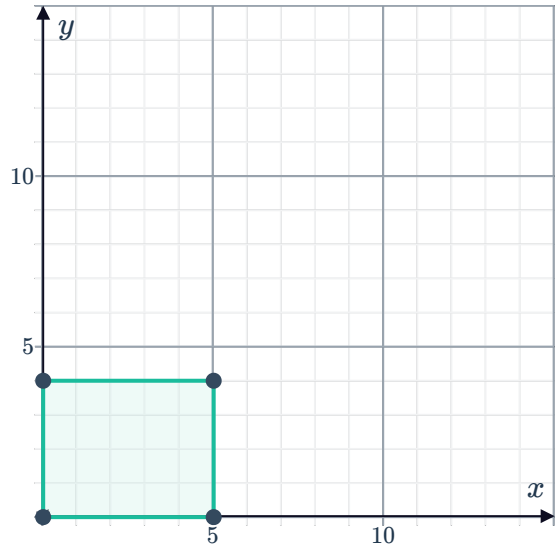
d



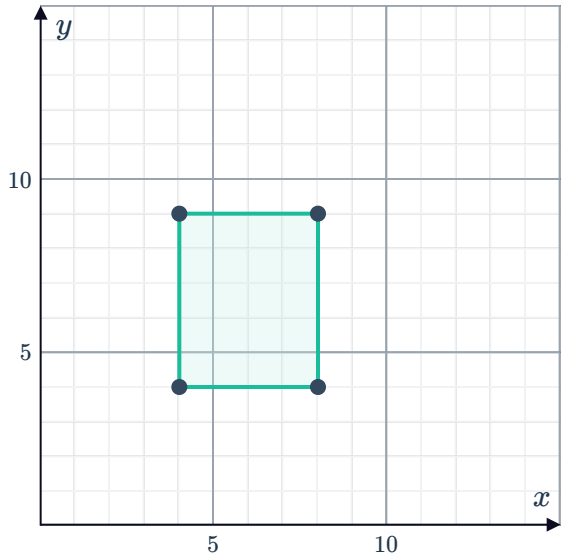
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2 Dilate each figure using the given scale factor and the origin as the center of dilation.

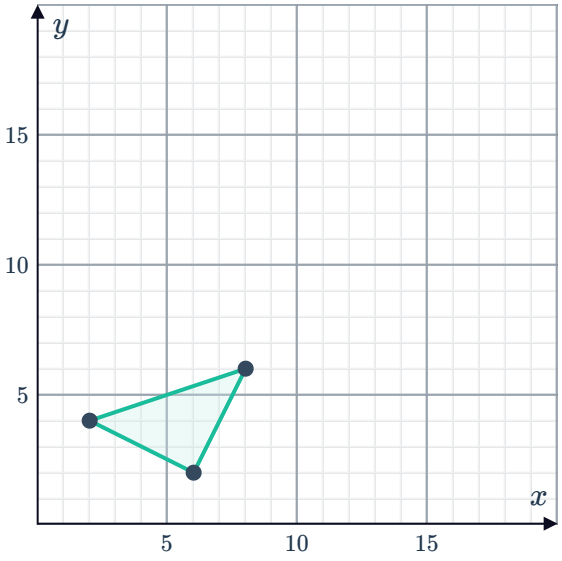
a Scale factor: 4



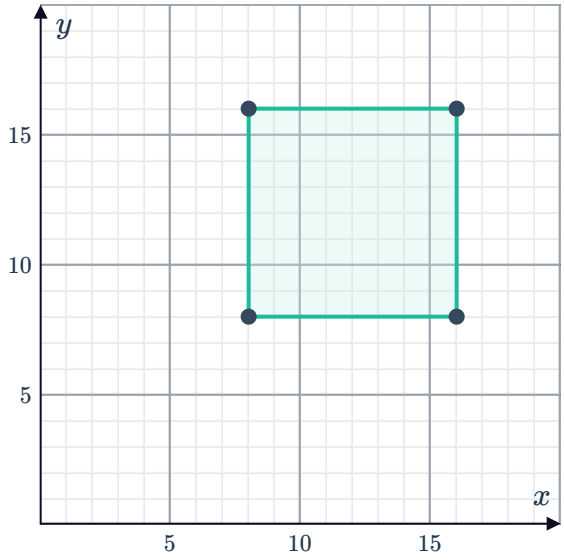
b Scale factor: 2



c Scale factor: 1.5

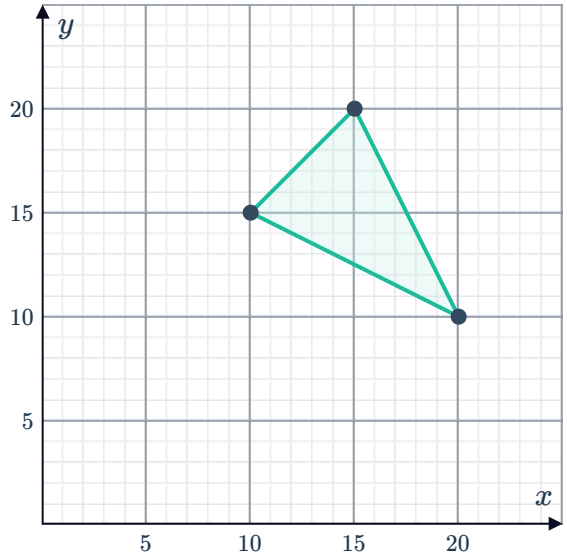


d Scale factor: $\frac{1}{2}$

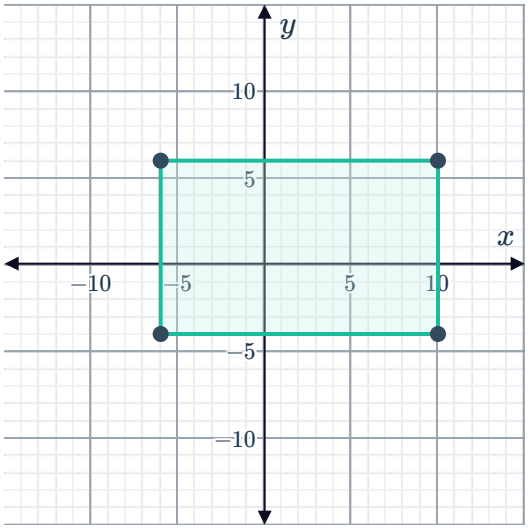


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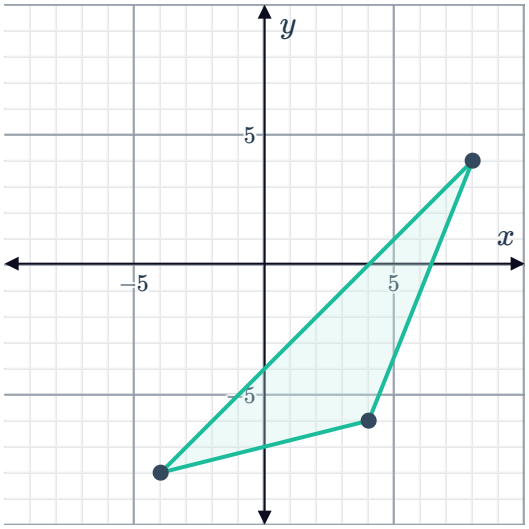
e Scale factor: $\frac{4}{5}$



f Scale factor: $\frac{1}{2}$



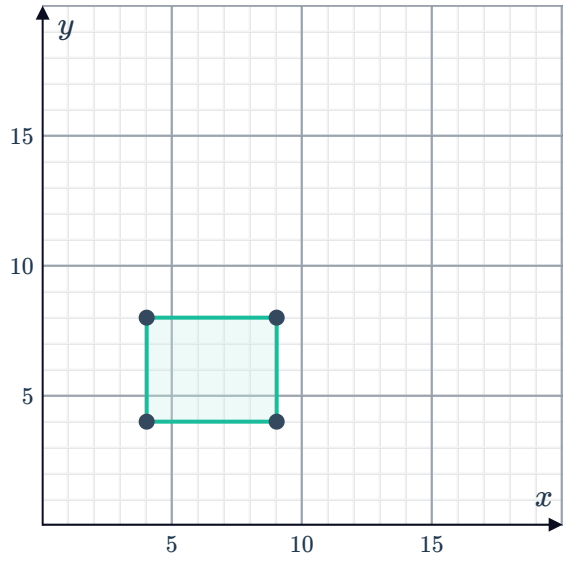
g Scale factor: $\frac{1}{2}$



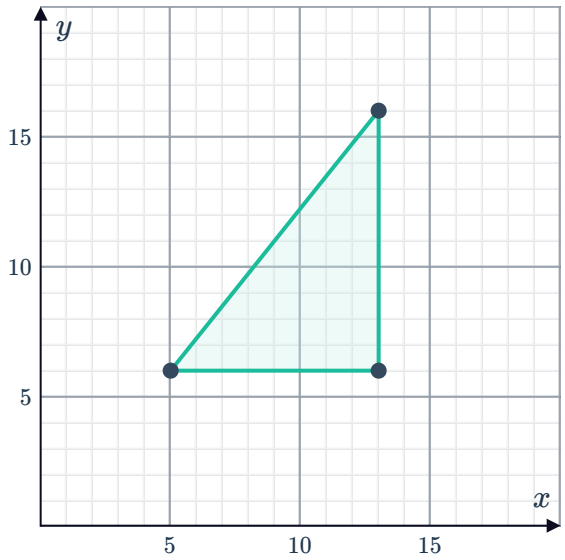
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3 Dilate each figure using the given scale factor and center of dilation.

a Scale factor: 2
Center of dilation: $A\,(2, 3)$



b Scale factor: $\frac{1}{2}$
Center of dilation: $A\,(3, 2)$



4 A triangle with vertices $A\,(-7, -5)$, $B\,(3, -5)$ and $C\,(-5, 9)$ is dilated by a factor of 0.5 using the origin as the center of dilation. State the coordinates of the vertices of the dilated triangle.

5 If a square with an area of $25\ \text{ft}^2$ is dilated by a factor of 0.4, find the side length of the dilated square.

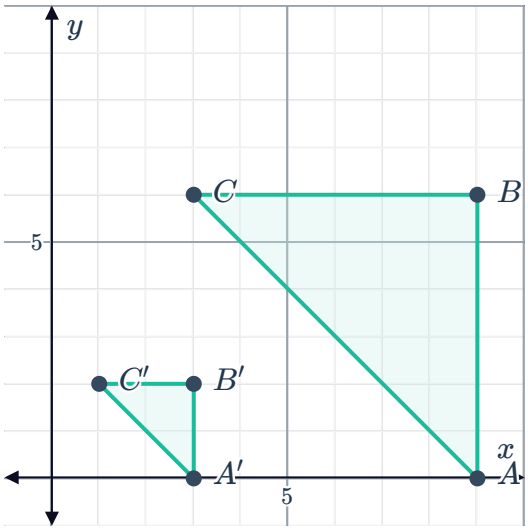
6 Find the scale factor of the following dilations:

a A rectangle with vertices $A\,(-5, 3)$, $B\,(3, 3)$, $C\,(3, -5)$ and $D\,(-5, -5)$ is dilated using the origin as the center of dilation. The vertices of the new rectangle are $A'\,(-25, 15)$, $B'\,(15, 15)$, $C'\,(15, -25)$ and $D'\,(-25, -25)$.

b A rectangle with vertices $A\,(-6, 9)$, $B\,(9, 9)$, $C\,(9, -12)$ and $D\,(-6, -12)$ is dilated using the origin as the center of dilation. The vertices of the new rectangle are $A'\,(-4, 6)$, $B'\,(6, 6)$, $C'\,(6, -8)$ and $D'\,(-4, -8)$.

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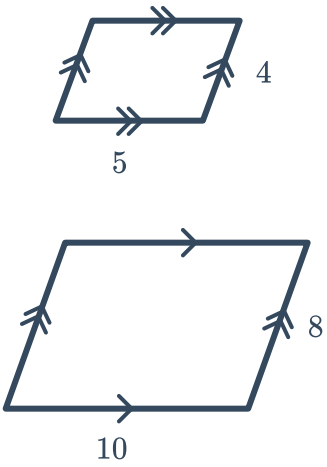
7 $\triangle ABC$ is dilated by a scale factor with a centre of dilation at the origin to obtain triangle $A'B'C'$.
Find the scale factor.



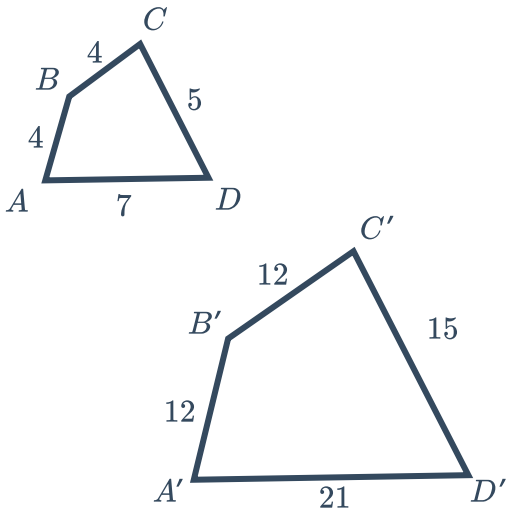
Additional questions

8 Find the scale factor for each pair of similar figures:

a

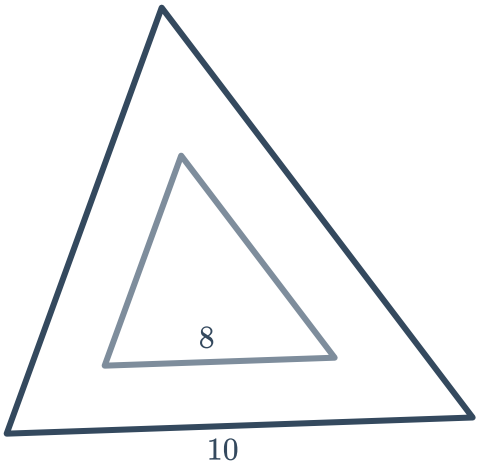


b

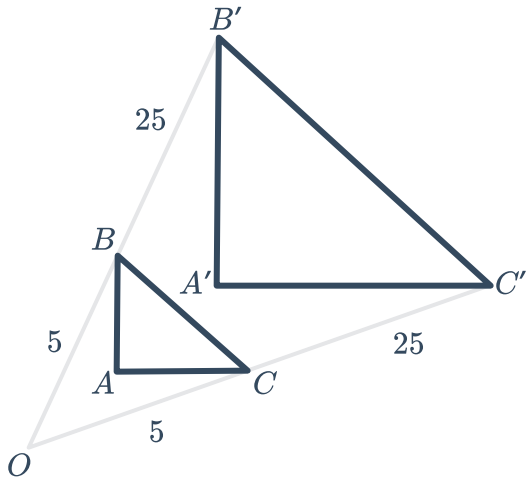


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c



d



9 Determine whether each statement is true or false. Justify your answer.

- **a** Any dilation with a scale factor less than 1 is a reduction.
- **b** Any dilation with a scale factor greater than 1 is an enlargement.
- **c** After dilating a figure, all corresponding angles will be congruent.

10 Layla describes the scale factor of a reduction as being in the form of $\frac{1}{k}$ since this is always a fraction less than one. Explain the error in her reasoning.

11 Use dilations to create a target: a bullseye made with four concentric circles where the width between rings is congruent. Create a diagram and justify your design.

12 Yousef needs to fit a **full** single sided sheet of notes on a single 3×5-inch notecard for his science test. Yousef's original notes are written on a standard 8.5×11-inch sheet of paper.

- **a** Determine the largest scale factor Yousef can use to make sure the original notes fit on the notecard.
- **b** Find the dimensions of Yousef's notes after applying the scale factor.
- **c** Yousef realizes he can't read any of his notes because the font is smaller than $\frac{1}{12}$ of an inch after dilation. Describe how Yousef can divide and dilate his sheet of paper to utilize both sides of his notecard and still have readable notes.

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